

Spring 2016

Problem 1 (Sp16). Show that

$$\int_4^9 \sqrt{-6 + 5\sqrt{-6 + 5\sqrt{-6 + 5\sqrt{-6 + 5\sqrt{x}}}}} dx$$

is a rational number.

Problem 2 (Sp16). Suppose that f and g are continuously differentiable real valued functions on $\mathbb{R}$ with $f, g, f', g' \in L^2(\mathbb{R})$. Show that

$$\int_{-\infty}^{\infty} f(x)g'(x) dx = - \int_{-\infty}^{\infty} f'(x)g(x) dx$$

Problem 3 (Sp16). Suppose g and f_n are nonnegative integrable functions such that $\int f_n \rightarrow 0$ as $n \rightarrow \infty$ and $f_n^2 \leq g$ for all n . Prove or find a counterexample to the statement that $\int f_n^4 \rightarrow 0$ as $n \rightarrow \infty$.

Problem 4 (Sp16). Prove that a monic polynomial $p(z)$ with real coefficients has all real zeros if and only if $\Im(p'(z)/p(z)) < 0$ whenever $\Im(z) > 0$.

Problem 5 (Fa87, Sp16). Evaluate the integral $I = \int_0^{2\pi} \frac{\cos^2 3\theta}{5 - 4 \cos 2\theta} d\theta$.

Problem 6 (Fa96, Sp16). Is there a real 2×2 matrix A such that, for $\varepsilon > 0$

$$A^{20} = \begin{pmatrix} -1 & 0 \\ 0 & -1 - \varepsilon \end{pmatrix}?$$

Exhibit such an A or prove there is none.

Problem 7 (Sp16). Suppose A is a symmetric matrix with rational entries and $A = UDU^T$, where U is orthogonal. Must D have rational entries? Prove or find a counterexample.

Problem 8 (Sp12, Sp16, Sp17). Find the structure of the group of units of $\mathbb{Z}_{2012}$.

Problem 9 (Sp16). Compute the Galois group of the normal closure of the field $\mathbb{Q}(\sqrt{3} + \sqrt{5})$ over $\mathbb{Q}$.

Problem 10 (Sp16). Show that $\int_0^\infty \frac{te^{-t/2}}{1-e^{-t}} = 4 \sum_{n=0}^\infty \frac{1}{(2n+1)^2}$.

Problem 11 (Sp16). Let $\{f_i\}$ and g be twice differentiable real valued functions on $\mathbb{R}$, with $f_i'' \geq 0$. Suppose that

$$\lim_{i \rightarrow \infty} f_i(x) = g(x)$$

for all $x \in \mathbb{R}$. Show that $g'' \geq 0$.

Problem 12 (Sp16). Show that the series

$$\sum_{k=1}^\infty \frac{(-1)^k}{k + |x|}$$

converges pointwise to a Lipschitz function $f(x)$. Is the convergence uniform on $\mathbb{R}$?

Problem 13 (Sp16). Compute

$$\int_C \frac{6z^5 + 1}{z^6 + z + 1} dz$$

where C is the circle centered at the origin with radius 2, oriented counter-clockwise.

Problem 14 (Sp16). Let $f(z) = \sum f_n z^n$ and $g(z) = \sum g_n z^n$ define holomorphic functions on a neighborhood of the closed unit disc $\overline{\mathbb{D}} = \{z \mid |z| \leq 1\}$. Prove that $h(z) = \sum f_n g_n z^n$ also defines a holomorphic function on a neighborhood of $\mathbb{D}$.

Problem 15 (Sp16). Let A be an $m \times n$ real matrix and $y \in \mathbb{R}^m$. Let $x \in \mathbb{R}^n$ be a vector with nonnegative entries that minimizes the euclidean distance $\|y - Ax\|$ (among all nonnegative vectors x). Show that the vector $v = A^T(y - Ax)$ has nonnegative entries.

Problem 16 (Sp16). Let A be a real square matrix and let ρ be the maximum of the absolute values of its eigenvalues i.e., its *spectral radius*.

- Show that if A is symmetric then $\|Ax\| \leq \rho \|x\|$ for all $x \in \mathbb{R}^n$, where $\|\cdot\|$ denotes the Euclidean norm.

- Is this true when A is not symmetric? Prove or give a counterexample.

Problem 17 (Sp16). Factor the polynomial

$$f(x) = 6x^5 + 3x^4 - 9x^3 + 15x^2 - 13x - 2$$

into a product of irreducible polynomials in the ring $\mathbb{Q}[x]$.

Problem 18 (Fa97, Sp09, Sp16, Sp26). Let G be a finite group.

1. If G is non-abelian, show that $G/Z(G)$ is not cyclic, where $Z(G)$ is the center of G .
2. If $|G| = p^n$, with p prime and $n > 0$, show that $Z(G)$ is not trivial and deduce that G has a normal subgroup of order p .
3. If $|G| = p^2$, show that G is abelian.